SENAY MESFIN

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Portfolio: ftd-minds.github.io/portfolio · GitHub: github.com/FTD-minds · Live demo: copilot-platform.onrender.com

PROFESSIONAL SUMMARY

Enterprise AI engineer with nine years across enterprise systems, ERP/accounting integrations, and AI-powered SaaS — with five live deployments, including three production AI systems. Built an enterprise generative-AI copilot platform with retrieval-augmented generation, source citations, and evaluation-gated CI/CD, productizing the exact integration failures I owned professionally for years. Equally at home presenting to executives and working inside the codebase. I help organizations solve real business problems by designing and building practical AI applications that simplify complex workflows, empower better decision-making, and create measurable business value.

SELECTED AI ENGINEERING PROJECTS

Enterprise AI Copilot Platform — Creator

Live: copilot-platform.onrender.com · Code: github.com/FTD-minds/copilot-platform

Open-source, model-agnostic platform for enterprise AI copilots — deployed and verified live. Module 1 diagnoses ERP sync and reconciliation failures modeled on problems I owned professionally for years. (Python · FastAPI · Docker · Postgres/pgvector · GitHub Actions)

- Implemented retrieval-augmented generation with source citations over a curated domain knowledge base, so every diagnosis is grounded and auditable rather than hallucinated.
- Built a model-agnostic LLM provider router (Anthropic/OpenAI, extensible to Bedrock/Vertex) — models swap by configuration, not code; a deterministic mock provider enables keyless, hermetic builds and a public demo with zero secrets in the image.
- Engineered a multi-step diagnostic agent (plan, act, synthesize) combining retrieval with computational tools (reconciliation calculator, mapping validator); tools additionally exposed over MCP.
- Established evaluation-gated CI/CD: a golden-case harness scores retrieval accuracy, tool selection, tool correctness, and groundedness on every push — quality regressions block merges; live public eval dashboard (87.5% / 100% / 100% / 100% across 8 golden cases); 20 tests; ruff + mypy; 6 ADRs.

MindNest (now ParentNest) — Sole Engineer

Live: mindnest-delta.vercel.app · Code: github.com/FTD-minds/mindnest-showcase

AI parenting companion wrapped in a complete subscription product — designed, built, and deployed end to end as sole engineer. (Next.js 14 · TypeScript · Supabase · Claude · Stripe · Vercel)

- Built the RAG pipeline (Voyage embeddings with pgvector cosine search) with reproducible per-user context assembly grounding every AI response.
- Security-first data layer: Postgres Row Level Security on every user-facing table, Supabase Auth, and route/admin gating in middleware.
- Production billing: Stripe checkout with a webhook handler as the sole writer of subscription state; CI gate runs lint, type-check, unit tests, an eval harness, and secret scanning.
- Public repository is a sanitized engineering showcase — proprietary prompts replaced with labeled placeholders; all engineering intact and runnable.

DialysisOS — Builder

AI-powered executive operations dashboard built for a dialysis-center operator; in use at one center. Demo linked from portfolio. (Claude API · Google APIs · OAuth)

- Claude API orchestration with Google API integrations (OAuth) powering natural-language operational reporting and business-intelligence views for leadership.
- Deliberate compliance boundary: business operations only — no HIPAA-covered patient data (PHI).

PROFESSIONAL EXPERIENCE

The Vitamin Shoppe — IT Support Engineer | 09/2021 – Present

- Selected for an internal AI initiative building a customer-facing chat assistant and operational dashboard for retail systems; authored the JSON configuration defining AI agent behavior and logic.
- Self-initiated, designed, and presented an interactive AI opportunity proposal to IT leadership (Director, Manager, and engineering team), mapping AI automation opportunities across service-desk, HR-system, and identity workflows.
- Key support resource for the enterprise finance-system migration (Sage Intacct to NetSuite) — validated data flow, configuration accuracy, and post-migration sync issues across the finance team's systems of record.

- Supported the rollout of Ramp as the AP automation platform as end user and informal liaison, identifying integration friction, configuration gaps, and workflow adjustments.
- Acted as the connecting point between finance stakeholders and technical teams — translating system-level issues into plain language and surfacing recurring problems before they became escalations.

Publishing.ai — Support Engineer / Solutions Advisor (AI-powered SaaS) | 08/2019 – 09/2021

- Diagnosed AI output inconsistencies, platform logic errors, and workflow failures in production for an AI-powered SaaS platform at scale, supporting high-value and Platinum-tier customers — hands-on AI product failure analysis that predates the LLM boom.
- Reproduced and isolated defects by validating inputs against expected behavior; filed engineering-grade bug reports (reproduction steps, logs, business impact); collaborated with Engineering through bug bashes, testing cycles, and release validation.
- Led live sessions demonstrating platform functionality during active bugs and beta launches — recreating reported issues live — and converted recurring issues into systemic stability, UX, and documentation recommendations; maintained the team's runbooks and escalation docs.

Shop Monkey — QuickBooks Online Integration Specialist | 06/2017 – 08/2019

- Selected for a newly created role as one of only two people company-wide owning the full QuickBooks Online integration experience — no support tier above the role except Engineering; final escalation point for QBO issues company-wide.
- Resolved the product's most common recurring failure pattern (item-level vs. chart-of-accounts mapping), a major driver of customer tickets, by walking customers through the actual sync logic live.
- Owned deposit and reconciliation troubleshooting end to end: bank-feed matching, undeposited funds, Stripe/ACH reconciliation, misapplied payments; handled Canada-specific (CRA) sales-tax sync in coordination with engineers.
- Reproduced every customer-reported bug in a sandbox before troubleshooting — rebuilding the exact scenario to isolate root cause rather than working from description alone.
- Tested new releases pre-rollout for integration impact; built the internal recorded-session training library used to onboard the next hire.

TECHNICAL SKILLS

- Languages: Python, TypeScript/JavaScript, SQL, HTML/CSS
- AI / LLM Engineering: generative AI, RAG (retrieval + source citations), evaluation harnesses and eval-gated CI, prompt engineering, agent orchestration, MCP, model-agnostic provider routing, Claude API, OpenAI API, embeddings and vector search (pgvector)
- Backend: FastAPI, Pydantic, REST API design, Next.js API routes, Stripe billing/webhooks, OAuth/SSO
- Frontend: Next.js 14, React, responsive and accessible UI (WCAG AA)
- Databases: PostgreSQL, pgvector, Supabase (Row Level Security, Auth, migrations)
- Deployment: Docker and docker-compose, GitHub Actions CI/CD, Render, Vercel, Netlify, GitHub Pages, secret hygiene (keyless hermetic builds, environment-injected credentials)
- Enterprise Systems: NetSuite, Sage Intacct, QuickBooks Online/Desktop, Ramp AP, ServiceNow, EasyVista, UKG, Active Directory, Okta, Google Workspace APIs
- Developer Tools: Git/GitHub, pytest, ruff, mypy, Architecture Decision Records, sandbox-first issue reproduction

ADDITIONAL PROJECTS

- Internal AI Platform — internal-ai-platform.netlify.app — self-initiated AI opportunity proposal presented to IT leadership; three interactive enterprise workflows (AI service-desk pipeline with human review and audit trail; operations intelligence; care assistant), every simulation explicitly labeled.
- FTD Quantitative Research — github.com/FTD-minds/ftd-quantitative-research — deterministic backtesting engine: byte-for-byte reproducible runs (SHA-256), out-of-sample methodology, 22-test dependency-free suite; reached and honored a documented negative result.

EDUCATION

University of Central Florida — Orlando, FL — Bachelor's degree